

Customer Churn Prediction Using Machine Learning

Bui Dinh Tuyen

June 13, 2026

Contents

1	Executive Summary	2
2	Business Understanding	2
3	Dataset Description	2
4	Data Warehouse Architecture	2
4.1	Bronze Layer	2
4.2	Silver Layer	3
4.3	Gold Layer	3
5	Feature Engineering	3
6	Exploratory Data Analysis	3
7	Machine Learning Models	4
8	Model Evaluation	5
9	Feature Importance Analysis	5
10	Business Insights	6
11	Power BI Dashboard	7
12	Power BI Dashboard	7
12.1	Customer Churn Overview	7
12.2	Customer Analysis	8
12.3	Machine Learning Insights	9
13	Technology Stack	9
14	Conclusion	9

1 Executive Summary

Customer churn is a critical challenge for telecommunication companies because customer acquisition costs are typically higher than customer retention costs.

This project develops an end-to-end churn prediction pipeline using SQL Server, Python, Machine Learning, and Power BI to identify customers who are likely to discontinue services.

The project achieved a ROC-AUC score of 0.851 using Logistic Regression and identified several important churn drivers, including monthly charges, tenure, contract type, and payment methods.

2 Business Understanding

The primary business objective is to reduce customer churn by identifying high-risk customers before they leave.

Key business questions:

- Which customers are most likely to churn?
- What factors influence churn behavior?
- How can the company improve retention strategies?

3 Dataset Description

The project uses the Telco Customer Churn dataset.

Metric		Value
Total Customers		7043
Features		23
Target Variable		churn_value
Positive Class		Churned Customer
Train Split		80%
Test Split		20%

Table 1: Dataset Summary

4 Data Warehouse Architecture

A Medallion Architecture was implemented using SQL Server.

4.1 Bronze Layer

Raw customer data was loaded directly from CSV files into SQL Server using BULK INSERT procedures.

4.2 Silver Layer

Data cleaning and standardization:

- Data type conversion
- Boolean transformation
- Missing value handling
- Data validation

4.3 Gold Layer

A curated feature-ready dataset was created for machine learning and business intelligence reporting.

5 Feature Engineering

The following preprocessing steps were applied:

- Selection of predictive features
- One-Hot Encoding for categorical variables
- Boolean conversion
- Train-test split

The final dataset contains 23 predictive features.

6 Exploratory Data Analysis

Several exploratory analyses were conducted to understand churn patterns.

Key findings include:

- Customers with shorter tenure are more likely to churn.
- Higher monthly charges are associated with increased churn.
- Fiber Optic customers show higher churn rates.
- Long-term contracts reduce churn probability.

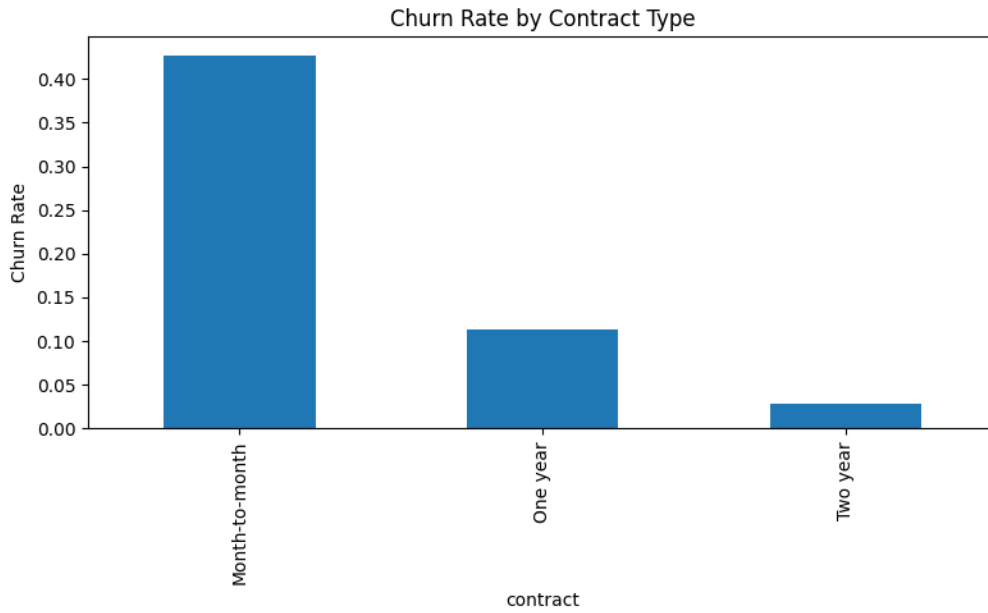


Figure 1: Feature Importance Analysis

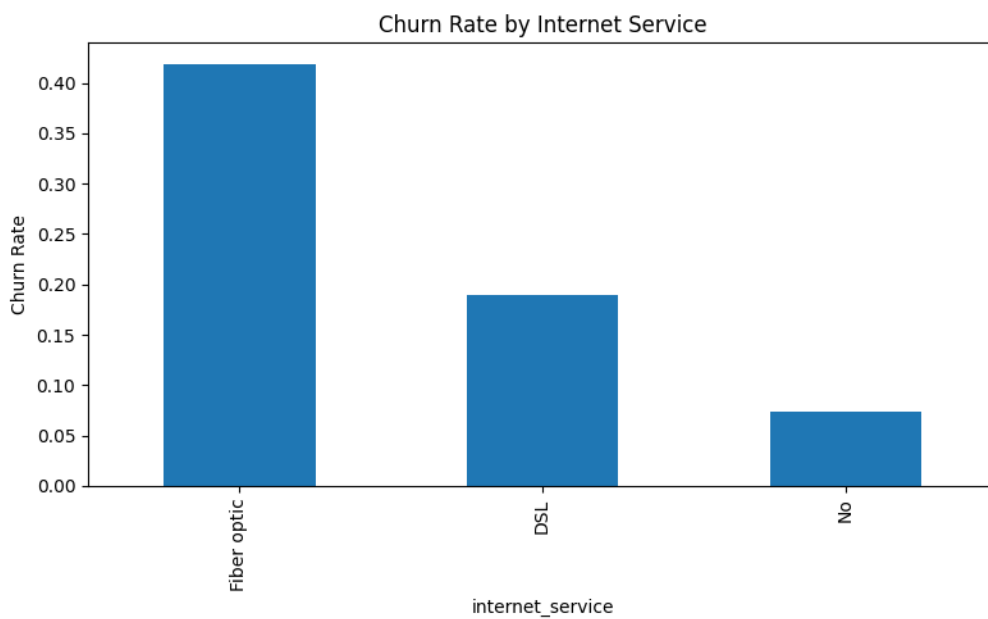


Figure 2: Feature Importance Analysis

7 Machine Learning Models

Three machine learning algorithms were evaluated:

1. Logistic Regression
2. Random Forest
3. XGBoost

8 Model Evaluation

Model performance was evaluated using ROC-AUC.

Model	ROC-AUC
Logistic Regression	0.851225
Random Forest	0.837131
XGBoost	0.831506

Table 2: Model Performance Comparison

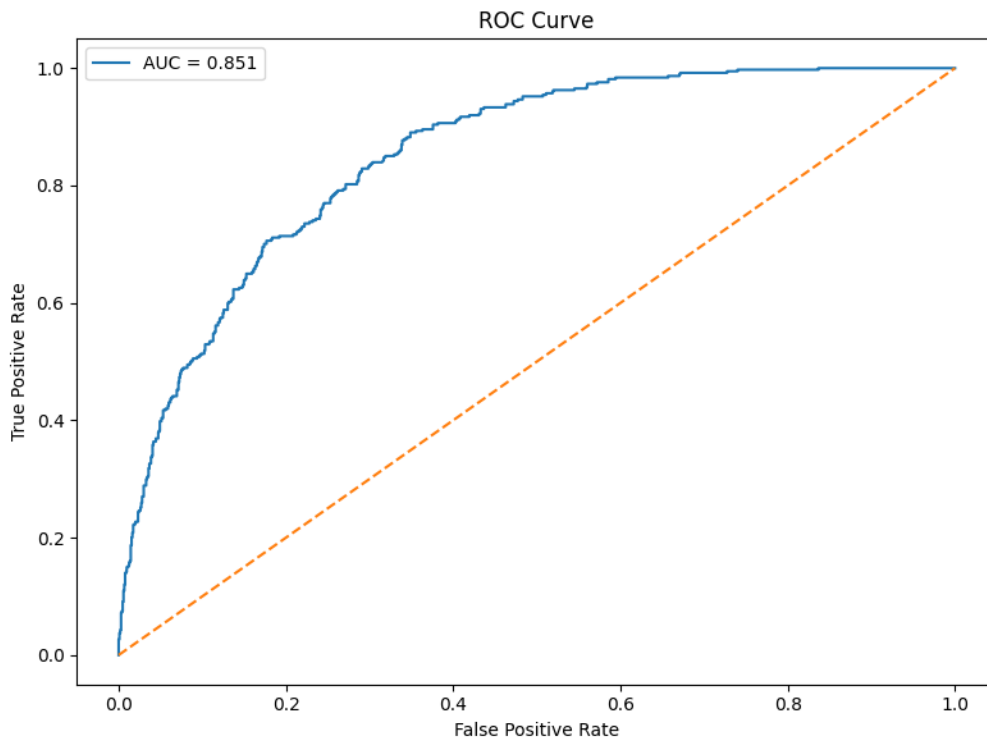


Figure 3: Feature Importance Analysis

Logistic Regression achieved the highest ROC-AUC score and was selected as the final model.

9 Feature Importance Analysis

Random Forest feature importance analysis identified the following top predictors.

Feature	Importance
total_charges	0.182148
monthly_charge	0.169667
tenure_months	0.169174
internet_service_Fiber optic	0.044103
payment_method_Electronic check	0.042055
dependents	0.040861
contract_Two year	0.033247
gender_Male	0.028102
paperless_billing	0.027041
partner	0.024723

Table 3: Top 10 Important Features

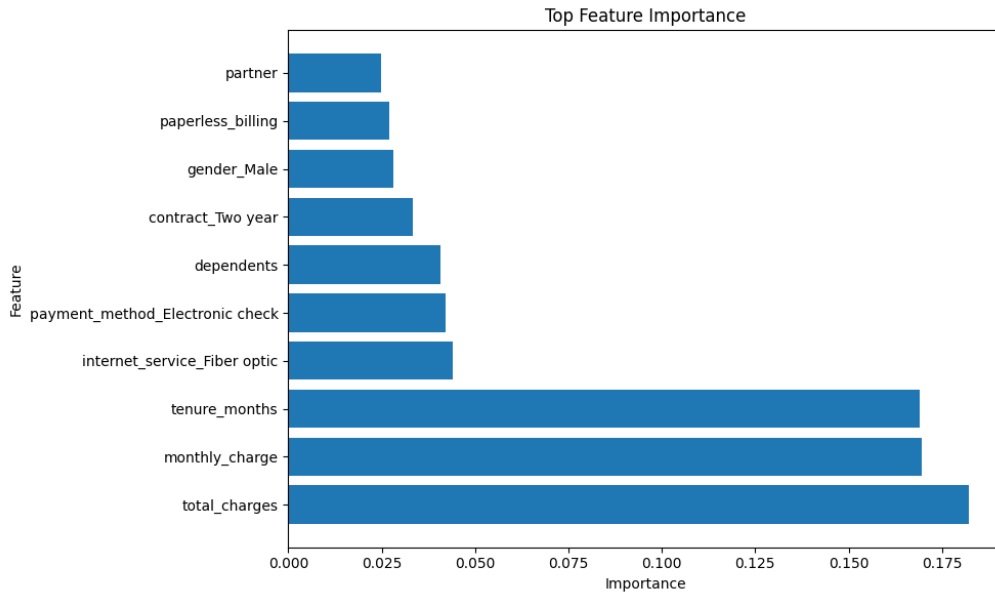


Figure 4: Feature Importance Analysis

10 Business Insights

Several important business insights were identified:

1. Customers with high monthly charges exhibit higher churn rates.
2. New customers are significantly more likely to leave.
3. Fiber Optic customers demonstrate elevated churn risk.
4. Electronic Check payment users show greater churn tendencies.
5. Long-term contracts help improve customer retention.

11 Power BI Dashboard

An interactive Power BI dashboard was developed to monitor customer churn and model performance.

Dashboard components include:

- Total Customers
- Churned Customers
- Churn Rate
- Average Monthly Charge
- Churn by Contract Type
- Churn by Internet Service
- Monthly Charge Distribution
- Machine Learning Performance Metrics

12 Power BI Dashboard

An interactive Power BI dashboard was developed to support business users in monitoring customer churn behavior and evaluating machine learning performance.

The dashboard consists of three main pages:

- Customer Churn Overview
- Customer Analysis
- Machine Learning Insights

12.1 Customer Churn Overview

This page provides a high-level summary of customer churn metrics, including total customers, churned customers, churn rate, average monthly charges, and churn distribution across contract and internet service types.

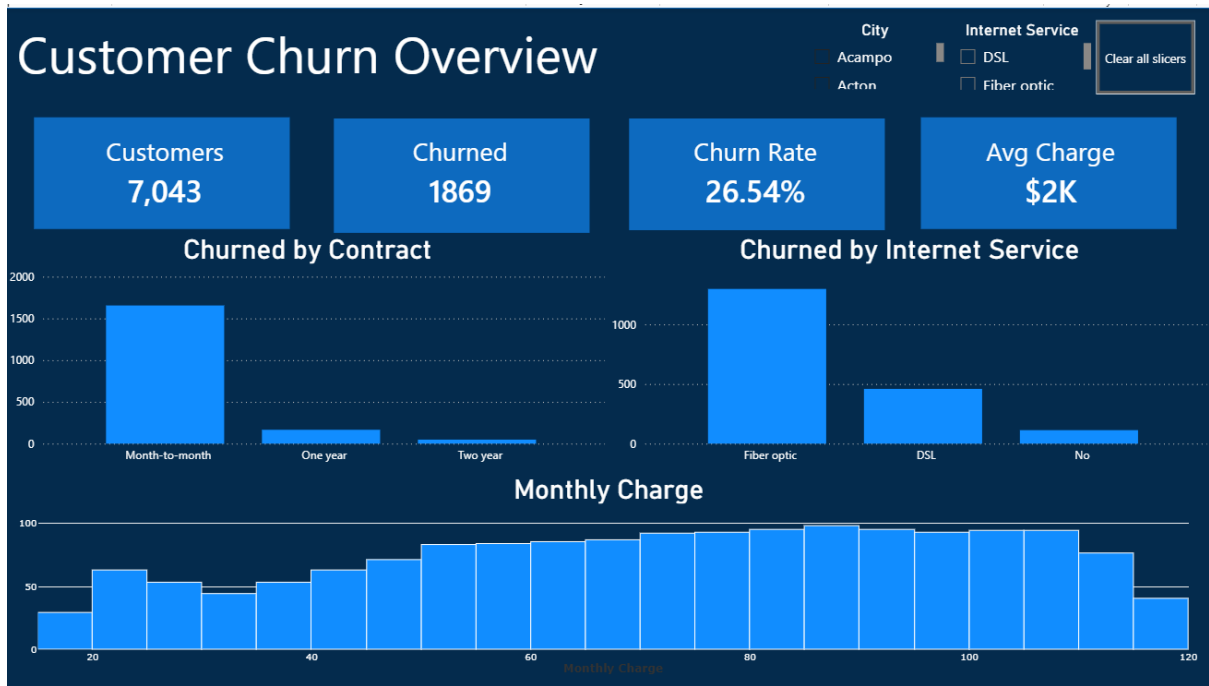


Figure 5: Customer Churn Overview Dashboard

12.2 Customer Analysis

This page focuses on customer characteristics and behavioral patterns associated with churn. Visualizations include monthly charges, tenure analysis, customer demographics, and service usage patterns.

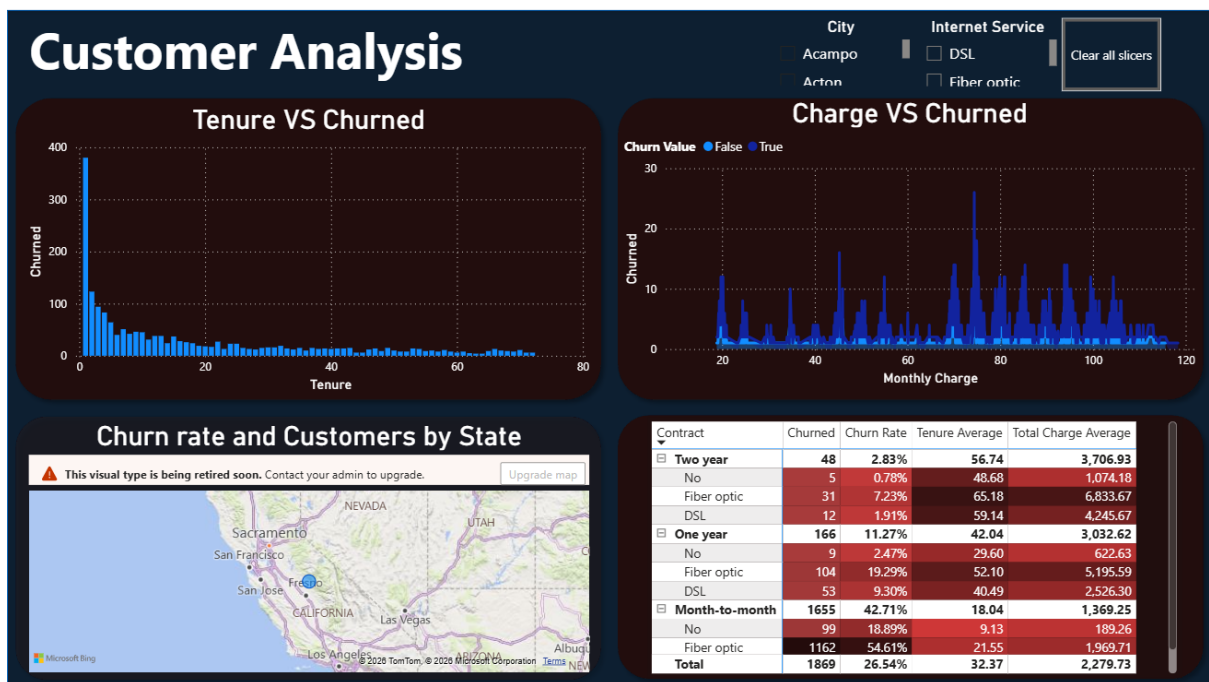


Figure 6: Customer Analysis Dashboard

12.3 Machine Learning Insights

This page presents model evaluation metrics and predictive insights, including ROC-AUC comparison, feature importance analysis, and model performance benchmarking.

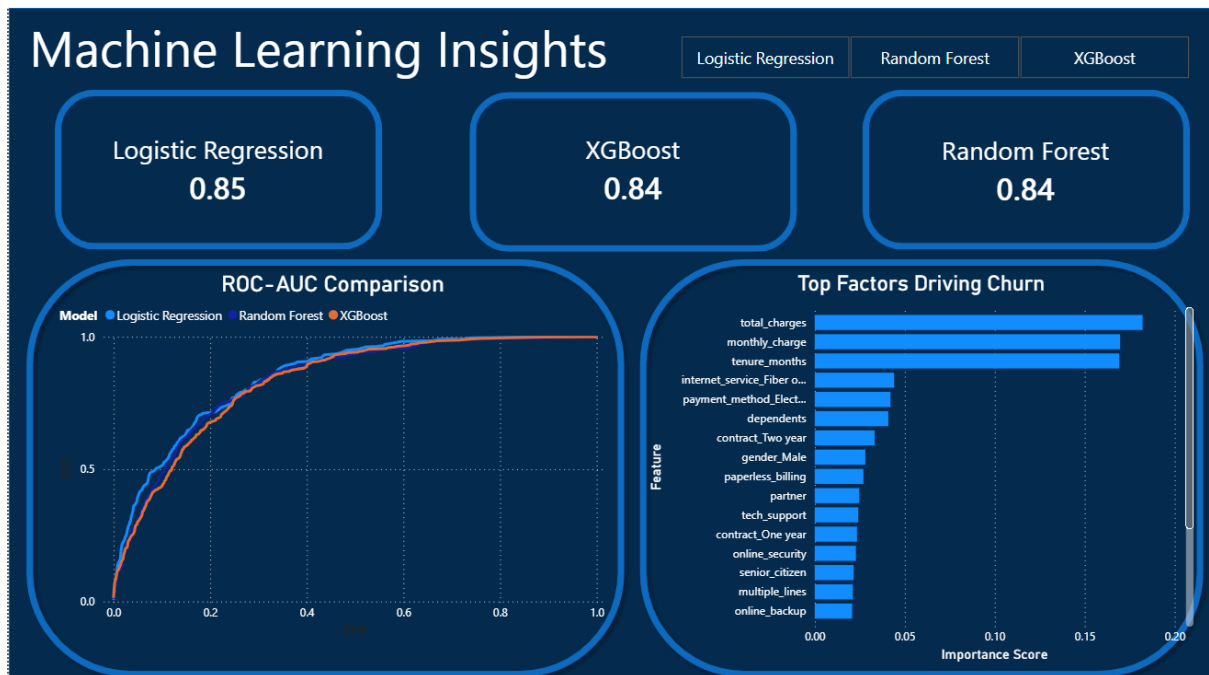


Figure 7: Machine Learning Insights Dashboard

13 Technology Stack

- SQL Server
- Python
- Pandas
- Scikit-Learn
- XGBoost
- Power BI
- GitHub

14 Conclusion

This project successfully implemented an end-to-end customer churn prediction solution, covering data warehousing, feature engineering, machine learning, and business intelligence reporting.

Among the evaluated models, Logistic Regression achieved the best predictive performance with a ROC-AUC score of 0.851225.

The analysis revealed that customer tenure, monthly charges, total charges, internet service type, and payment methods are the most influential factors affecting churn.

These findings can support proactive retention strategies and improve customer lifetime value.